

Econ 30065: Microeconomics

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Externalities II

Example

Assume the paper market is perfectly competitive and that paper production generates harmful water pollution. Let's assume for now that it is not possible to reduce pollution.

There are 200 paper mills whose total cost of production is given by $TC_i(Q) = 500Q_i + Q_i^2$. The External cost of a mill's production is given by $EC_i = 100Q_i + Q_i^2$.

The demand curve is given by $Q^D = 150000 - 100P$.

We want to find the competitive equilibrium and the efficient one, and see what is the size of inefficiency (DWL) caused by the paper production not taking into account the externality.

Solution

➤ Competitive equilibrium:

1. $P=MC$ for each firm

➤ $MC=500+2Q_i$

➤ $P=500+2Q_i$

➔ $Q_i=0.5P-250$

2. $Q^S=200Q_i=100P-50000$

3. Set $Q^D=Q^S$

➤ $150000-100P=100P-50000 \rightarrow P=1000, Q=50000$

Solution

➤ Socially efficient equilibrium:

1. $P = MSC$

➤ $MEC_i = 100 + 2Q_i$

➤ $MSC_i = MC + MEC = 500 + 2Q_i + 100 + 2Q_i = 600 + 4Q_i$

➤ $Q_i = 0.25P - 150$

2. $Q^s = 200Q_i = 50P - 30000$

3. $Q^D = Q^s$

➤ $150000 - 100P = 50P - 30000 \rightarrow P = 1200, Q = 30000$

Solution

To find the inefficiency we need to find the area of the deadweight loss

$$DWL = 0.5(Q^{\text{comp}} - Q^{\text{eff}})(P^{\text{eff}}(Q^{\text{comp}}) - P^{\text{comp}})$$

In our case:

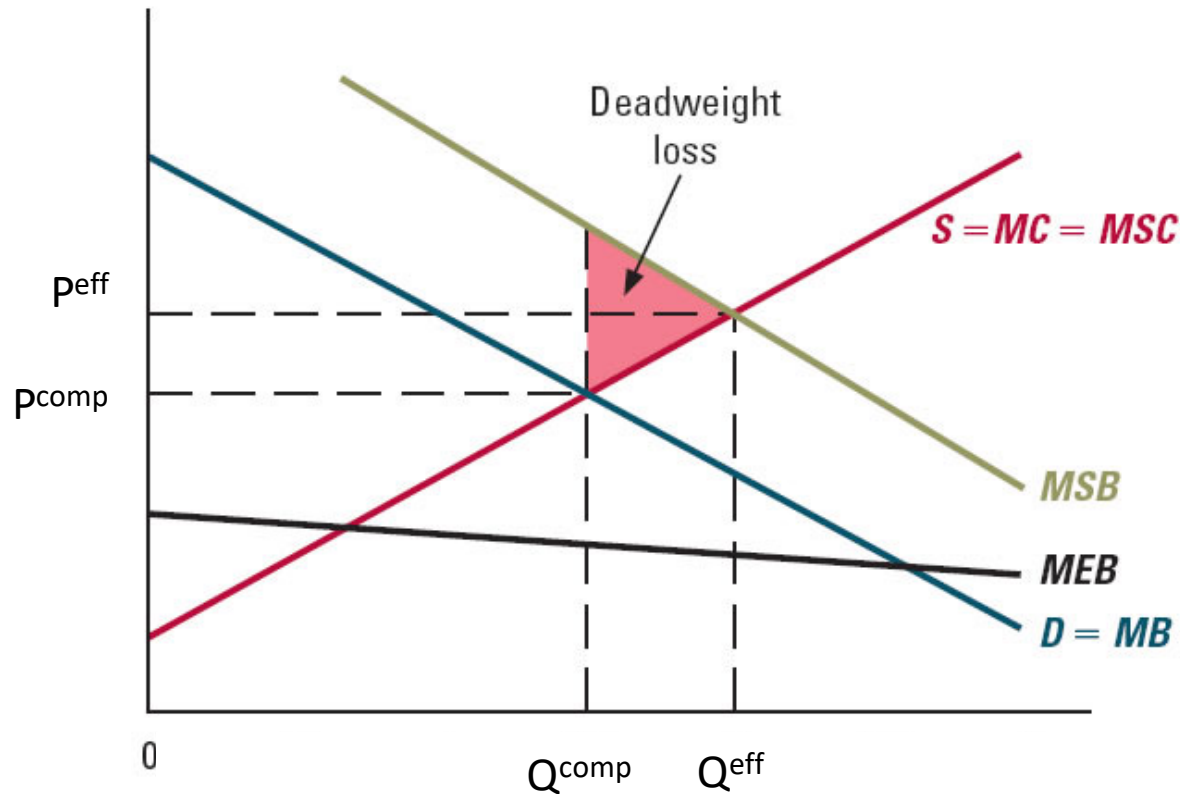
$$P^{\text{Eff}}(Q^{\text{comp}}) = Q/50 + 600 = 1600$$

$$DWL = 0.5(50000 - 30000)(1600 - 1000) = 6.000.000$$

Positive Externalities and Inefficiency in competitive markets

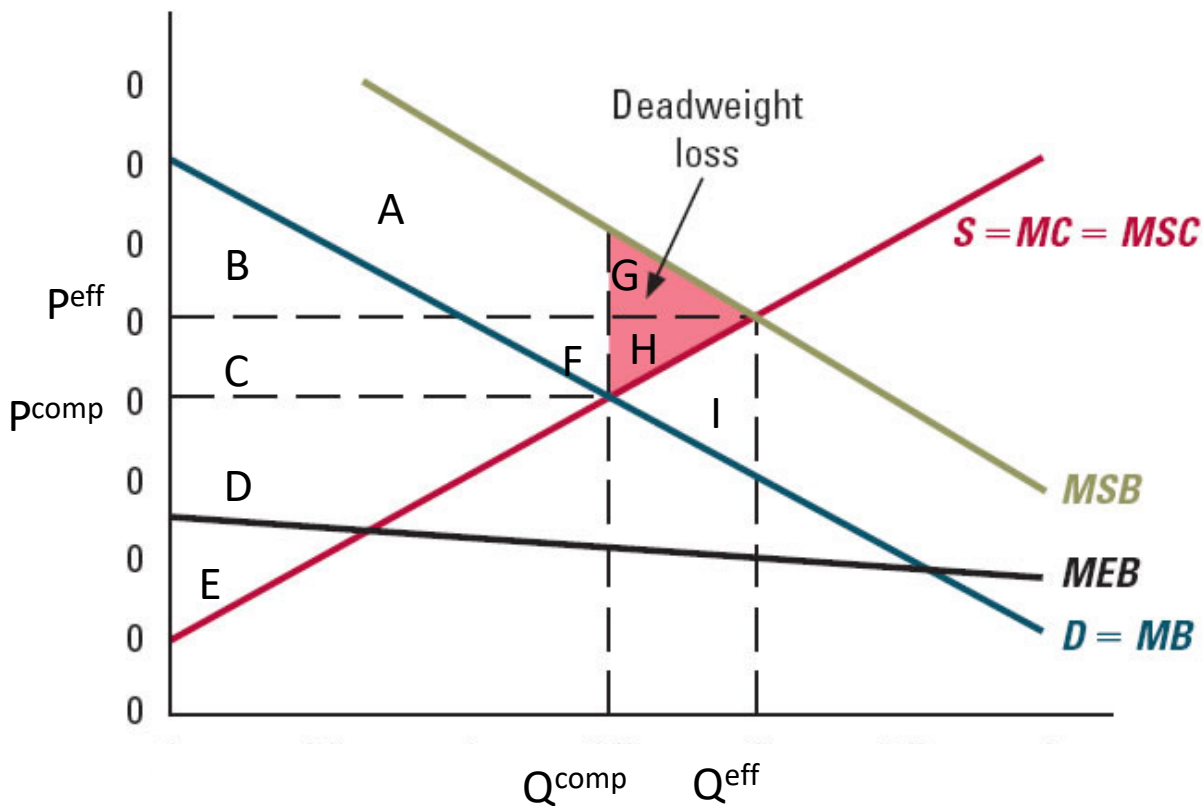
- Competitive equilibria are inefficient also when positive externalities are present
 - An **external benefit** $EB > 0$ is the economic gain that a positive externality provides to others
 - MEB is the marginal external benefit.
- Optimality condition in perfect competition $P(=MB)=MC$
- With positive externality $MSB=MB+MEB=P+MEB$
 - ➔ With positive externality social efficiency condition should be:
 $MSB=MC$
- Firm competitive output will be inefficiently low from society's perspective: since $MB=MC \rightarrow MSB > MC$
- Like a negative externality, a positive externality creates deadweight loss

Positive Externality in a Competitive Market



Positive Externality in a Competitive Market

	Perfect competition	Efficient level of output
CS	B+C	B-F-H-I
PS	D+E	C+D+E+F+H
External Benefit	A+F	A+F+G+H+I
Social welfare	A+B+C+D+E+F	A+B+C+D+E+F+G+H



Externalities and inefficiency of competitive markets

- For negative externalities:
 - Without externalities, we can clearly say that competitive equilibrium is more efficient than monopoly
 - With externalities this does not need to hold:
 - Example: negative externality → in perfect competition firms produce too much
 - We know monopolist produces less than perfect competition: it could reduce inefficiency of perfect competition (or not)
 - ➔ We cannot say which one is less efficient, unless we have data!
- For positive externalities it is still true that monopoly is less efficient than perfect competition

Remedies for Externalities:

Private Sector

- Let's focus on negative externalities (same holds for positive externalities)
- Whenever the allocation of resources is inefficient we now it is possible to arrange mutually beneficial trades
- Private parties have incentives to identify inefficiencies and negotiate solutions: this will increase their surplus

PRIVATE NEGOTIATION

Example: Negotiation

- Factory earns \$600,000 in profits and it pollutes a river; to abate pollution, it can buy a water purifier at cost $C=100,000$
- A farmer uses the water for irrigation
 - He makes \$370,000 if water clean and \$250,000 if water polluted
- Joint profits:
 - 850,000 if factory pollutes
 - $970,000 - C$ if factory does not pollute

Example: Negotiation

- It is efficient to invest in water purifier (total surplus would increase)
 - If the factory has the right to pollute the river
 - Then the farmer will pay for the water purifier, because that would increase his profits: $370,000 - 100,000 > 250,000$. The factory has same profits as before: 600,000.
 - Pareto improvement!
 - No pollution
 - If the farmer has the right, the company will pay for the purifier, and has lower profits of 500,000
- If abatement of pollution is efficient, market will negotiate to eliminate it!

Property rights

- Who has the property rights does not influence the level of pollution but influences only the profits!
 - If it is efficient not to have pollution, no matter who has the property rights, pollution disappears
 - If it is efficient not to abate pollution, no matter who has the property rights, pollution does not disappear
- **Coase theorem:** If bargaining is frictionless, regardless of how property rights are assigned, voluntary agreements will restore efficiency

Limitations of Bargaining

- In our example the farmer and the factory are creating a new market: the market for exchanging a right (to pollute)
 - They remedy the externality by creating a new market
- Every externality can be traced to missing markets
- Private negotiations lead to transactions that the parties would have made if the required markets were not missing
- Problem: many factors that cause externalities also make bargaining fail
 - Bargaining can be impractical
 - Difficult logistics, substantial time and effort (e.g. air pollution caused by traffic)
 - Property rights may be ambiguous (who has the rights to the river, the farmer or the factory?) ➔ legal battles
 - Limited information can lead to an impasse (parties may have an incentive to over/under state the costs of the externality)
 - Contracts may be difficult to enforce

Public sector intervention

- When private negotiations fail to address market failures associated with externalities, government policies can potentially improve **social efficiency**
 - **Social efficiency:** the socially efficient outcome maximizes the total surplus, no matter whether one party will be worse off (with respect to perfect competition)
 - Socially efficient outcome is Pareto efficient, but it may not be a Pareto improvement over perfect competition outcome
- 1. Policies that support markets (e.g., establishing clear property rights)
- 2. Quantity controls (e.g., emissions standards)
- 3. Policies that correct private incentives (e.g., taxes, fees, subsidies)

Policies that support a market

- In some situations, governments can address externalities by helping the private sector create the necessary markets
 - Establish clear property rights, pass laws that protect those rights, and enforce contracts
 - **Property rights:** enforceable claim on a good or resource
- Sometimes market remedies are not possible

Quantity Controls

- Government can attempt to address an externality by regulating the activity that produces it
- An ***emissions standard*** is a legal limit on the amount of pollution that a person or company can produce when engaged in a particular activity

Policies that Correct Private Incentives

- Some policies address externalities by forcing people to internalize external costs and benefits
 - Impose taxes or fees, provide subsidies (Pigouvian remedies)
- **Pigouvian taxation** involves the use of taxes or fees to remedy negative externalities
- A Pigouvian tax forces a decision-maker to internalize the marginal external costs associated with her activity
 - Set tax equal to MEC(at Q efficient) → $MC=MSC$!
- Ideal Pigouvian tax reproduces the price that the good in question would command in an efficient competitive equilibrium